LAB EXPERIMENTS

Q20) Write a C program for ECB mode, if there is an error in a block of the transmitted ciphertext, only the corresponding plaintext block is affected. However, in the CBC mode, this error propagates. For example, an error in the transmitted C1 obviously corrupts P1 and P2.

a. Are any blocks beyond P2 affected?

b. Suppose that there is a bit error in the source version of P1. Through how many ciphertext blocks is this error propagated? What is the effect at the receiver?

Aim:

To write a C program to demonstrate the effect of transmission errors in **ECB (Electronic Codebook)** and **CBC (Cipher Block Chaining)** modes and analyze error propagation.

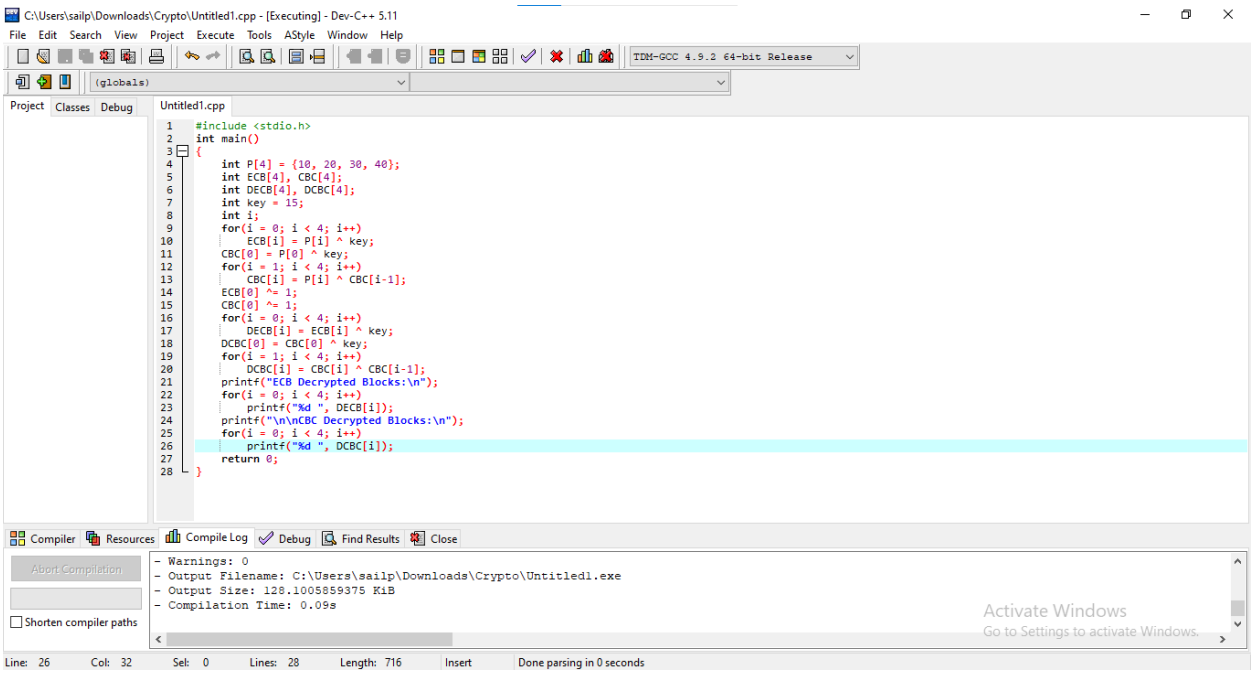
Algorithm:

1. Start the program.
2. Initialize four plaintext blocks and a secret key.
3. Encrypt each block independently for **ECB mode** using the XOR operation.
4. Encrypt each block for **CBC mode**, where each plaintext block is XORed with the previous ciphertext block before encryption.
5. Introduce a transmission error (bit flip) in the first ciphertext block.
6. Decrypt the modified ciphertext in **ECB mode** and observe the affected plaintext blocks.
7. Decrypt the modified ciphertext in **CBC mode** and observe the affected plaintext blocks.
8. Display the decrypted plaintext for both modes.
9. Compare the error propagation in ECB and CBC.
10. Stop the program.

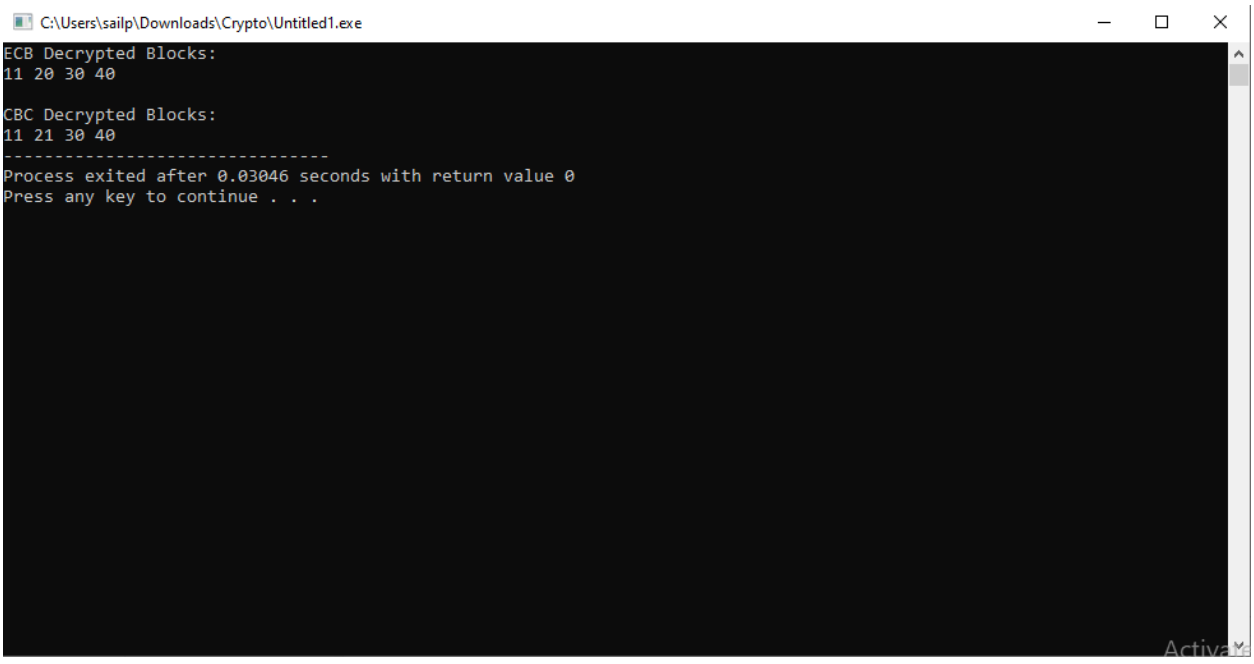
Result:

Thus, the C program demonstrating **ECB** and **CBC** modes was successfully executed. It was observed that **ECB affects only the corresponding plaintext block**, whereas **CBC propagates the error to the next plaintext block as well.

Code:


C:\Users\sailp\Downloads\Crypto\Untitled1.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
Untitled1.cpp
#include <stdio.h>
int main()
{
int P[4] = {10, 20, 30, 40};
int ECB[4], CBC[4];
int DECB[4], DCBC[4];
int key = 15;
int i;
for(i = 0; i < 4; i++)
ECB[i] = P[i] ^ key;
CBC[0] = P[0] ^ key;
for(i = 1; i < 4; i++)
CBC[i] = P[i] ^ CBC[i-1];
ECB[0] ^= 1;
CBC[0] ^= 1;
for(i = 0; i < 4; i++)
DECB[i] = ECB[i] ^ key;
DCBC[0] = CBC[0] ^ key;
for(i = 1; i < 4; i++)
DCBC[i] = CBC[i] ^ CBC[i-1];
printf("ECB Decrypted Blocks:\n");
for(i = 0; i < 4; i++)
printf("%d ", DECB[i]);
printf("\n\nCBC Decrypted Blocks:\n");
for(i = 0; i < 4; i++)
printf("%d ", DCBC[i]);
return 0;
}
Compiler Resources Compile Log Debug Find Results Close
Abort Compilation
Shorten compiler paths
- Warnings: 0
- Output Filename: C:\Users\sailp\Downloads\Crypto\Untitled1.exe
- Output Size: 128.1005859375 KiB
- Compilation Time: 0.09s
Activate Windows
Go to Settings to activate Windows.
Line: 26 Col: 32 Sel: 0 Lines: 28 Length: 716 Insert Done parsing in 0 seconds

Output:


C:\Users\sailp\Downloads\Crypto\Untitled1.exe
ECB Decrypted Blocks:
11 20 30 40
CBC Decrypted Blocks:
11 21 30 40

Process exited after 0.03046 seconds with return value 0
Press any key to continue . . .